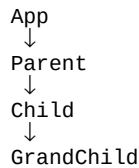


Props Drilling in React - Complete Notes

Props Drilling happens when data is passed from one component to another through multiple levels of components, even when intermediate components do not need that data.

1. What is Props Drilling?

Suppose Parent Component has data and Child Component needs it. But between them there are many intermediate components. The data is passed through every component using props. This process is called Props Drilling.



2. Problem with Props Drilling

- Code becomes difficult to manage
- Intermediate components receive unnecessary props
- Difficult to maintain large applications
- Makes debugging harder

3. Example of Props Drilling

```
// App.jsx
import Parent from "../Parent";

function App() {
  const name = "Vishvesh";
  return <Parent name={name} />;
}

export default App;

// Parent.jsx
import Child from "../Child";

function Parent({ name }) {
  return <Child name={name} />;
}

export default Parent;

// Child.jsx
import GrandChild from "../GrandChild";

function Child({ name }) {
  return <GrandChild name={name} />;
}

export default Child;

// GrandChild.jsx
```

```
function GrandChild({ name }) {  
  return <h1>Hello {name}</h1>;  
}  
  
export default GrandChild;
```

4. Solution to Props Drilling

The most common solution is React Context API. Context API allows components to access data directly without passing props manually at every level.

5. Context API Flow

Context Provider
↓
Any Component can access data directly

6. Benefits of Context API

Benefit	Meaning
Less Props Passing	Avoids unnecessary props
Cleaner Code	Components become simpler
Easy State Sharing	Data accessible everywhere
Better Maintainability	Improves large applications

7. Important Interview Definition

Props Drilling is the process of passing data through multiple nested components using props, even when intermediate components do not use that data.